

Traverse & Network Adjustment in Trimble Business Center

A step-by-step Parametrix guide to preparing survey data in TBC, choosing the right adjustment, and running either a traverse adjustment (compass or transit rule) or a least-squares network adjustment, then reviewing, documenting, and signing off the results.

Level Field and office survey staff

Software Trimble Business Center

Time about 60 minutes

01 • OVERVIEW

Before you start

An adjustment reconciles the small measurement errors in a survey with the control you hold, so the final coordinates are consistent with that control and with each other. This guide covers the two ways to do that in TBC:

- **Adjust Traverse** creates a conventional traverse, reports its angular and linear closure, and distributes the closure through the traverse with the Compass (Bowditch) or Transit rule. ^[1] ^[4]
- **Adjust Network** runs a **least-squares** adjustment of all the enabled observations in the network, weighted by their expected precision, and reports statistics that show how well they agree. ^[9]

Sections 01–05 apply to both. Section 06 explains how to choose between them. Then follow either the traverse workflow (07–09) or the network workflow (10–15). The two methods are alternatives, not two steps of one process.

Closed loop

Starts and ends on the same known point, with a known backsight azimuth. Gives a check on angles and distances, but cannot detect a scale or azimuth error in the starting control.

Closed connecting

Starts on one known pair of points and ends on a different known pair. Preferred, because it checks against independent control.

Open traverse

Does not close on known control. You can't compute a misclosure, so there is nothing to adjust. Avoid open traverses for control work.

You will need

- Trimble Access job files (.job or JobXML .jxl) or other supported raw data from the data collector.
- Published coordinates for at least the start point and backsight, and for the closing point and foresight on a connecting traverse.
- The project's datum, coordinate system, and units, plus the accuracy standard the job must meet (contract, agency such as WSDOT, or ALTA/NSPS). ^[13] ^[14]
- For a network adjustment: any GNSS vectors, level runs, or other observations that tie into the same points.
- A TBC installation appropriately licensed for the survey and adjustment commands used here. Trimble's help doesn't tie these commands to a specific license tier, so check with your CAD/software administrator.

VERSION NOTE

Menu locations below follow the ribbon layout in current TBC releases. If your version looks different, search TBC for the command name, for example "Adjust Traverse."

Superscript numbers such as ^[1] link to the [References](#). Steps with a Trimble citation were checked against Trimble's online TBC help. Steps without one are standard office practice, so check them in your copy of TBC.

† A menu path, button, field, or option that is *not* confirmed by the cited TBC help. It's shown as a typical location; verify it in your TBC version.

■ 02 · PROJECT

Project setup

1 Create a new project

Choose **File > New Project** [†] and pick the Parametrix survey template if one exists for your office. Otherwise use the default metric or US survey foot template.

2 Set the coordinate system

Go to **Project Settings > Coordinate System** [†]. Select the zone, for example *NAD83(2011) / Washington North (US ft)*, and the geoid model the project calls for.

3 Check units and the grid/ground setting

In **Project Settings > Units** [†], confirm distance (US survey ft or m), angle (DMS), and display precision. If the project is on ground coordinates, confirm the combined scale factor or project-to-ground settings before importing anything.

4

Set computation options

Under **Project Settings > Computations**, confirm whether distances are reduced to grid or ellipsoid, and whether curvature and refraction corrections are applied.[†] The Computations settings page itself is documented in Trimble's help.^[5]

5

Set the default standard errors

Trimble says to check **Project Settings > Default Standard Errors > Total Station** before importing total station data. That's also where you can set default standard, centering, and height errors.^[20] There is a separate GNSS section.^[16] For total station data, TBC takes its starting values from the imported job file unless you choose otherwise.^[16] They set how each observation is weighted in a network adjustment, so they must be right before you run one. Never leave applicable standard errors at 0.00. See [Network: setup](#) for what they mean and the Parametrix values.

6

Save the project

Save it to the project folder using Parametrix's standard naming convention before you import data.

WATCH OUT

The wrong vertical units or grid/ground setting is the most common cause of a traverse that “closes badly” when the field work was actually fine. Check these settings before you import.

03 · DATA

Import field data

1

Open the Import pane

Home > Import[†]. You can also drag and drop .jxl files straight into the open project.[†]

2

Browse to the data folder

Select the .job or .jxl files and click **Import**[†]. If TBC warns about a coordinate system mismatch, click **Details**[†] to see a side-by-side comparison of the project and job coordinate systems. Keep the *project* coordinate system unless you know the field job is correct.

3

Check the import

In the **Project Explorer**, expand *Station Setups*[†] and *Points*. Each traverse station should have one setup (or more if you observed it in multiple sessions), with the correct backsight.

Check the shape in Plan View

This is a quick visual check for blunders before you go any further. Look for:

- **A station or line that sticks out** from where it should be. This usually means a wrong point ID, the wrong backsight, or a typo in a height or distance.
- **A station plotted at or near 0,0**. TBC had no starting coordinates for it, so it couldn't be computed.
- **A loop that doesn't come back to its start**. On a closed loop, the last leg should end very close to the starting point.

04 • CONTROL

Set control points

The adjustment holds the control you choose and adjusts everything else to fit. TBC needs to know which coordinates you want to hold.

1

Enter or import control coordinates

Use the **Create Point**[†] command, or import a control CSV.

2

Set the coordinate quality to Control

Select the point in the Project Explorer, open its **Properties** pane, and set the coordinate quality to **Control** (horizontal, and vertical if you are holding elevation). The quality levels, from highest to lowest, are *Control*, *Survey*, *Mapping*, and *Unknown*.^[6]^[7] To change several points at once, Ctrl-select them in the Project Explorer and change the quality in Properties.^[8] In a network adjustment, Control quality alone doesn't hold a point; you choose which to fix (see [Network: minimally constrained](#)).

3

Match the point IDs

Control IDs must match exactly the IDs used in the field (for example CP1, not CP-1). If they don't, rename the points or merge the duplicates.

4

Verify the backsight azimuth

Compute an inverse between the start point and the backsight (the **Inverse**[†] command) and compare it to the published azimuth or bearing.

FIXED VS. ADJUSTED END STATIONS

TBC holds a start or end station **fixed** when its position and orientation are known without the rest of the traverse, such as a setup over a known point or a resection. A start or end station positioned only from the traverse itself is **adjusted**. If both ends are that type, only the backsight and foresight are known, and Trimble calls this the weakest kind of traverse.^[2]

Review raw observations

Fix blunders *before* you adjust. An adjustment spreads error around; it can't fix a bad observation.

CHECK	WHERE IN TBC [†]	WHAT TO LOOK FOR
Instrument & target heights	Station Setup properties; observation properties	Missing or 0.000 HI/HT, or a rod height that doesn't match the field notes
Prism constant	Observation properties › Target	Mixed constants (for example 0 mm vs –30 mm) between setups
Rounds / direct & reverse	Station setup observations in the Project Explorer	Large residuals in the mean turned angle; rounds outside spec
Backsight check	Station Setup › Backsight observation	Backsight distance or elevation residual well beyond the instrument's accuracy
Duplicate observations	Project Explorer › Points	The same point shot from both directions; reciprocal distances should agree within a few hundredths
Atmospheric / PPM	Station Setup properties	Temperature and pressure entered (or 0 ppm) applied the same way at every setup

To disable a bad observation, select it and disable it in its properties[†]. Don't delete it, because you want an audit trail.

■ 06 · CHOOSE A METHOD

Choose the adjustment method

Both commands produce adjusted coordinates, but they answer different questions and are used on different kinds of surveys.

Adjust Traverse

For a conventional traverse where you want to review the **angular and linear closure** and then distribute that closure through the traverse stations with the Compass (Bowditch) or Transit rule. ^{[1] [4]}

Adjust Network

A **least-squares** adjustment. Use it when there is meaningful **redundancy**, such as cross-ties, repeated observations, multiple loops, GNSS vectors, conventional data tied to GNSS control, leveling, or other interconnected observations. ^{[9] [12]}

	ADJUST TRAVERSE	ADJUST NETWORK
TBC command location	Survey > Optical > Adjust Traverse ^[1]	Survey > Network > Adjust Network ^[9]
Adjustment method	Angular misclosure distributed first; then linear misclosure distributed by the Compass (Bowditch) or Transit rule ^{[3] [4]}	Least squares: all enabled observations adjusted together, each weighted by its standard error ^{[9] [12]}
Type of survey or network	A single conventional traverse (closed loop or closed connecting) between known start and end stations ^[2]	Redundant networks: cross-ties, repeated observations, multiple loops, GNSS, conventional plus GNSS control, leveling
How closure or residuals are evaluated	Angular misclosure, linear misclosure, and precision ratio, before and after adjustment ^[3]	Residuals, standardized residuals and Tau outliers, Reference Factor, Chi Square test, degrees of freedom, error ellipses, control coordinate comparisons ^{[17] [18]}
Report reviewed	Traverse Adjustment Report ^[3]	Network Adjustment Report ^[17]
How results are stored	Traverse-adjusted coordinates on the traverse stations ^[5]	New coordinate records with quality <i>Adjusted</i> or <i>Fixed in network adjustment</i> , promoted to final. ^[9] Any traverse adjustment is cleared and the raw traverse observations are used. ^[21]
When to use	A simple traverse with no meaningful redundancy, or when a specification or client requires closure and precision ratio	Any survey with meaningful redundancy or mixed observation types, or when statistical QA of the observations is needed
Warning	Do not use Adjust Traverse and Adjust Network as competing final adjustments on the same traverse. Choose one method for the final solution, and report only that one.	

ONE FINAL METHOD PER TRAVERSE

You may review a traverse's closure with Adjust Traverse first, as a QA check, and keep that preliminary report with the QA records. But if Adjust Network is selected as the final method, the **network adjustment becomes the final coordinate solution**. When Adjust Network is run on a survey that contains traverse-adjusted coordinates, TBC clears the traverse adjustment and uses the raw traverse observations in the least-squares adjustment. ^[21] TBC does not keep two competing final adjusted coordinate sets, so don't present the traverse adjustment as a final result once the network adjustment has been run.

Words that mean different things

Control (quality)	A coordinate quality level in TBC, the highest in the hierarchy (Control, Survey, Mapping, Unknown). ^[6] ^[7] It identifies trusted coordinates. In a network adjustment, Control quality alone doesn't hold a point; you choose which control coordinates to fix. ^[9]
Fixed (traverse)	A start or end station whose position and orientation are known independently of the traverse. The traverse adjustment doesn't change it. ^[2]
Fixed (network)	A control coordinate you check as fixed (2D , h , or e) in Adjust Network. TBC holds it by giving it a very small standard error (default 0.000001 m). ^[9] ^[19] You can instead enter standard errors from the control's data sheet so the point is held only as tightly as it's known. ^[9]
Adjusted	In a traverse adjustment, the coordinates written to the traverse stations. ^[5] In a network adjustment, a coordinate quality label (<i>Adjusted</i> or <i>Fixed in network adjustment</i>) on the new records the adjustment creates. ^[9]

■ 07 · TRAVERSE ADJUSTMENT

Traverse: create the traverse

TBC has no separate “define” step. You create, edit, and adjust a traverse in the same **Adjust Traverse** command pane. ^[1] If the traverse was already adjusted in Trimble Access, it comes in with the job, and you can use the same command to edit the station list or settings and re-adjust it. ^[1]

1

Check the traverse project settings

Open **Project Settings > Computations > Traverse**. ^[1] Here you choose whether traverses are adjusted automatically or manually. In manual mode, TBC re-checks the traverse on every project compute but doesn't change the adjusted coordinates. It flags the traverse for re-adjustment when the change exceeds the horizontal and vertical tolerances you set. ^[5]

2

Open Adjust Traverse

Survey > Optical > Adjust Traverse ^[1]

3

Name and create the traverse

In **Traverse name**, enter a name that includes the job and date, for example PX-2601_Loop1_2026-10-08, then click **Create**. ^[1]

4

Select stations in order

Add the station setups in the order they were occupied: the start station, the intermediate stations, then the end station. Each intermediate station must have a foresight to the next station. ^[2]

5

Set the adjustment options

Choose the angle, horizontal, and elevation methods described in the [next section](#).

6

Preview Results

Click **Preview Results** to see the closure before anything is written to the project. ^[1] The Traverse Adjustment Report's **Before Adjustment** section shows the same misclosures, precision values, and traverse length. ^[3]

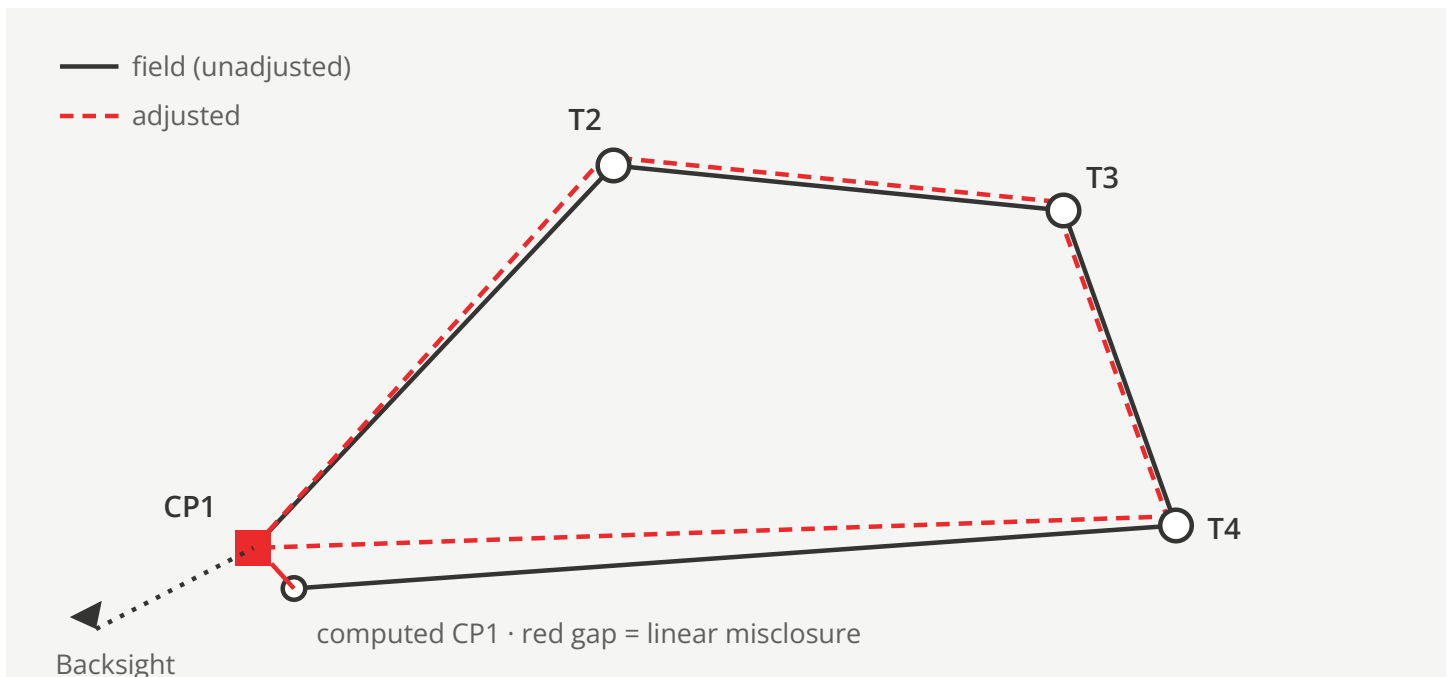


Figure 1: In a closed loop, the computed coordinates of the closing point miss the true control position. That vector is the linear misclosure. (Not to scale; error exaggerated.)

STOP AND CHECK

Compare the angular and linear closure against the applicable project, agency, client, or office standard. Where the governing specification sets the angular misclosure limit as $K \cdot \sqrt{n}$ seconds (n = number of angles), K comes from that specification. ^[11] If the closure exceeds the allowable value, or the precision ratio is below spec, go back to [Review raw observations](#). Don't adjust a traverse that fails closure.

08 • TRAVERSE ADJUSTMENT

Traverse: adjust & review

TBC applies the angular correction first and then distributes the linear misclosure; the Traverse Adjustment Report shows the results in that order. ^[3] ^[4]

1 Choose the angle adjustment

Under **Adjust angles**, TBC offers **Proportional to Distance**, **Equal Proportions**, or **None**. ^[1] Trimble describes them this way: ^[1]

- **Proportional to Distance**: adjustments are weighted more heavily for angles with longer traverse lines than for angles with shorter traverse lines.
- **Equal Proportions**: adjustments are weighted equally, regardless of the length of the traverse lines.
- **None**: no angle adjustment.

2 Choose the horizontal adjustment

Under **Adjust horizontal**, TBC offers **Transit**, **Compass/Bowditch**, or **None**. ^[1]

- **Compass (Bowditch)**: adjusts latitudes and departures in proportion to the length of each leg. ^[4] Trimble notes that it assumes distances and directions are measured with consistent precision. ^[1]
- **Transit**: adjusts latitudes and departures in equal proportions or in proportion to the lengths of the lines. ^[4] Trimble notes that it assumes directions are measured with higher precision than distances. ^[1] Textbooks reserve it for cases where angles are much more precise than distances. ^[11] ^[15]

3 Choose the elevation adjustment

Under **Adjust elevation**, TBC offers the same three options: **Proportional to Distance**, **Equal Proportions**, or **None**. ^[1] Choose **None** if you are carrying elevations from a separate level loop.

4 Apply

When the preview is within spec, click **Apply**. ^[1] TBC writes adjusted coordinates to the traverse stations. Fixed start and end stations are not changed. ^[2]

5

Open the Traverse Adjustment Report

Click the **Traverse Adjustment Report** icon on the Adjust Traverse pane toolbar, or open it from the Reports list in the Quick Access Toolbar. ^[3] Review these sections:

- **Before Adjustment:** raw misclosures, precision, and length
- **After Angular Adjustment** and **After Distance Adjustment:** what's left after each correction
- **End-Point Orientations** and **Fixed points:** confirm TBC held the control you expected
- **Station Adjustments:** the correction applied to each station, with a plan diagram ^[3]

ADJUST BEFORE YOU BUILD

Topo shots are computed from their station's coordinates, so adjust the traverse *before* you build surfaces or linework, and check a few sideshots afterward. Also note that merging a point in a manually adjusted traverse removes all of its traverse-adjusted coordinates and its report. ^[5]

09 · TRAVERSE ADJUSTMENT

Traverse: worked example (compass rule)

This is a four-sided closed loop starting and ending at **CP1** (N 5000.000, E 5000.000), in US survey feet. The azimuths shown have already been angularly balanced.

LEG	AZIMUTH	DIST (FT)	LATITUDE	DEPARTURE
CP1 → T2	45°00'00"	412.38	+291.597	+291.597
T2 → T3	135°00'04"	385.09	-272.305	+272.294
T3 → T4	224°59'57"	412.33	-291.566	-291.557
T4 → CP1	315°00'05"	385.16	+272.356	-272.343
Σ		1594.96	+0.082	-0.009

$$\text{Linear misclosure} = \sqrt{(0.082^2 + 0.009^2)} = 0.082 \text{ ft}$$

$$\text{Precision ratio} = 1594.96 / 0.082 \approx 1 : 19,400$$

$$\text{Compass correction (leg } i\text{): } C_{\text{lat}} = -\Sigma\text{Lat} \times (D_i / \Sigma D) \quad \cdot \quad C_{\text{dep}} = -\Sigma\text{Dep} \times (D_i / \Sigma D)$$

STATION	C _{LAT}	C _{DEP}	ADJUSTED N	ADJUSTED E
T2	-0.021	+0.002	5291.576	5291.599
T3	-0.020	+0.002	5019.251	5563.895
T4	-0.021	+0.002	4727.664	5272.341
CP1 (close)	-0.020	+0.002	5000.000	5000.000

The adjusted traverse closes exactly on CP1. A compass adjustment of the same observations in TBC, with the same settings, should agree with these values within displayed rounding; compare them against the report's **Station Adjustments** table. ^[3] The method follows standard textbook practice. ^[11] ^[15] Working one traverse by hand like this is a good way to confirm you understand what the software is doing.

■ 10 · NETWORK ADJUSTMENT

Network: setup

A least-squares adjustment uses every enabled observation and weights each one by how precise you say it is. ^[9] ^[12] ^[22] The settings below control that weighting, so check them before every network adjustment.

1 Review the network adjustment settings

Open **Project Settings** from the Quick Access Toolbar and select **Network Adjustment** in the left pane. ^[19] Note the confidence level used to display error ellipses and precisions (1-sigma, 95%, or 99%), so you know what the report's numbers represent, and the default standard error used for fixed control coordinates. ^[19] Leave the other settings at their defaults unless your survey lead says otherwise.

2

Verify the default standard errors

Open **Project Settings > Default Standard Errors** and check the **Total Station** and **GNSS** sections. ^[16] ^[20] Confirm:

- The angle and distance standard errors match your instrument, for example 1" or 2" for angles and 1 mm + 1.5 ppm for EDM. For total station data these may come from the imported job file. ^[16]
- The **centering** and **height** (measure-up) errors for the instrument and target. ^[20]
- The GNSS standard errors, including centering and antenna height. ^[16]
- Standard errors for any other observation types in the network, such as leveling.

PARAMETRIX PRACTICE (INTERNAL VALUE)

Parametrix's working value is **0.01 ft** for centering and **0.01 ft** for measure-up, for both total station and GNSS. This is an internal value, **not a Trimble value**, and it is not yet backed by an approved Parametrix standard. Confirm it with the survey lead for each project. Trimble's help confirms that centering and height errors are set in Default Standard Errors; ^[20] the exact field labels in your version aren't confirmed here. [†]

3

Understand what the standard errors do

A standard error is the **expected precision** of an observation: how far off you'd expect it to be, typically. Least squares gives more weight to observations with small standard errors and less to those with large ones. ^[12] If they're too small (optimistic), good observations look like failures; if too large (pessimistic), real problems can hide.

4

Never leave applicable standard errors at 0.00

A zero standard error tells the adjustment an observation is perfect. That causes major problems in the adjustment and in its statistics. Pay special attention to GNSS and to centering and height errors.

5

Open Adjust Network

Survey > Network > Adjust Network . ^[9]

■ 11 · NETWORK ADJUSTMENT

Network: minimally constrained adjustment

Minimally constrained = **Does my survey agree with itself?**

Trimble's workflow starts with a minimally constrained adjustment, which applies only the constraints needed to define the coordinate system, so you can evaluate the **internal consistency** of the survey. ^[9] With only the minimum held, the control can't hide or distort problems in the observations, and a bad observation can't be blamed on the control. Fix the observations first; test them against the project control later.

1

Hold only the minimum control

On the constraints tab of the Adjust Network pane, check **2D**, **h** (ellipsoid height), or **e** (elevation) for the coordinates you want to hold. ^[9] Trimble's workflow help calls it the **Constrained Coordinates** tab; some versions label it **Constraints** or **Fixed Coordinates**. ^[9] Hold only what's needed to position and orient the network. For a GNSS network that's usually one point held in 2D and height. ^[12] A conventional-only network also needs its orientation defined, for example by a known azimuth. ^[12]

TBC REQUIREMENT

Adjust Network requires at least **one horizontal fix** and **one vertical fix**. They don't need to be on the same point. ^[9] ^[18] An elevation (**e**) can be fixed for a GNSS point only when the project's coordinate system has a geoid; otherwise, fix the ellipsoid height (**h**). ^[18]

2

Run the adjustment

Click **Adjust**. ^[9] The **Results** tab shows a summary of the adjustment. ^[9]

3

Review the results

Open the **Network Adjustment Report** from the pane toolbar ^[9] and review the items below. See [Network: read the report](#) for where each one appears.

What the numbers mean

Residual	How much the adjustment had to change an observation to make everything fit. ^[17]
Standardized residual	A normalized measure of the residual relative to its expected uncertainty. It accounts for both the observation's standard error and the geometry of the network, so angles, distances, GNSS vectors, and level observations can be compared statistically on the same scale. The larger it is, the less the observation agrees with the rest of the network than its precision says it should. ^[12] TBC uses standardized residuals with the Tau criterion to identify potential outliers. ^[17] ^[18]
Tau outliers	Observations whose standardized residual fails the Tau test. TBC shows them in red and treats them as outliers that you must examine and justify keeping. ^[17] ^[18]
Reference Factor	One overall number comparing how well the observations actually fit with how well their standard errors said they would. About 1.00 means they agree. ^[17] Above 1 means the observations fit worse than expected (blunders, or standard errors that are too optimistic). Below 1 means they fit better than expected (standard errors that are too pessimistic). ^[12]

Chi Square test

A statistical test at 95% of whether the Reference Factor is consistent with 1.00 for the amount of redundancy in the network. ^[17] ^[18] It can fail high or low. It's a test of the observations *and* the assumed standard errors together, not of the observations alone.

Degrees of freedom

The number of observations beyond the minimum needed to compute the points: the network's **redundancy**. ^[18] **Degrees of freedom = 0 means there is no redundancy.** Every observation is needed just to compute the points, so the observations aren't independently checking one another. The adjustment can fit them exactly, and the residuals, Reference Factor, and Chi Square test can't reveal a blunder. Low degrees of freedom also make the statistics weak. ^[12]

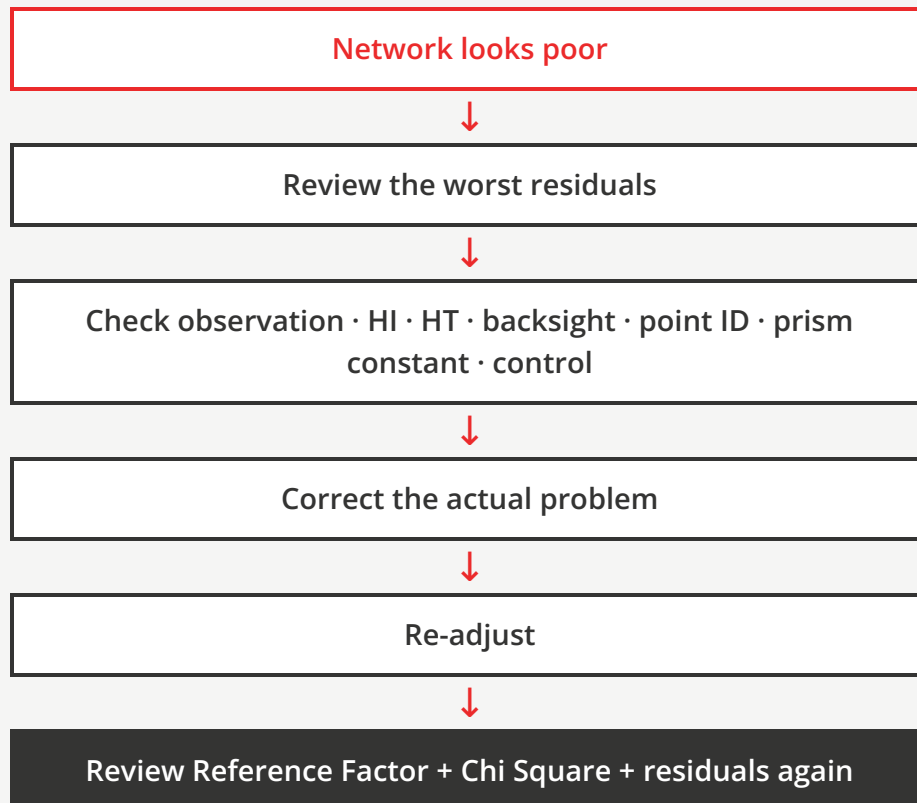
Error ellipses

The size and orientation of each point's estimated uncertainty, at the confidence level set in Network Adjustment settings. ^[17] ^[19] Large or very elongated ellipses mean weak geometry, such as a point tied in by only one observation.

■ 12 · NETWORK ADJUSTMENT

Network: troubleshooting

Least squares spreads a blunder into the observations around it, so the observation with the biggest residual isn't always the bad one. Work systematically, change one thing at a time, and keep notes.



Still poor? Go back to the top with the new worst residual.

1

Work from the largest standardized residuals down

The report lists observations with the worst standardized residuals at the top, and Tau outliers in red. ^[17] Start there.

2

Look for the cause

Check the instrument height, target height, backsight, point ID, prism constant, the observation's quality (rounds, residuals, field notes), and that control points are the right points. Also look for other likely field or office blunders, such as a mis-keyed value, a duplicate point name, or a wrong coordinate system or geoid.

3

Fix the actual cause where possible

Correct the height, ID, or setting in the data and re-adjust. A corrected observation is always better than a removed one.

4

Disable only with a documented reason

If an observation is clearly bad and can't be corrected, disable it; Adjust Network uses only enabled observations. ^[9] Record which observation, why, and who decided. Don't disable observations just because they're red or to make the test pass.

5

Re-run after each meaningful change

Re-adjust and check that the statistics and residuals changed for the reason you expected before moving to the next problem.

6

Leave the weighting alone unless there's a reason

Don't change weights simply to force the Chi Square test to pass. A passing test should come from sound observations and realistic standard errors. If the observations look sound, with no outliers and agreement with independent checks, but the assumed standard errors clearly don't reflect how the survey actually performed, the weighting may be revised. Trimble's process uses the **Weighting** tab to improve the a priori error estimates, with the goal of bringing the Reference Factor toward 1.00. ^[9] Document the reason. **Reset to 1** returns all scalars to 1.00. ^[9]

CHI SQUARE ACCEPTANCE

A passing Chi Square test is the normal Parametrix acceptance target, consistent with Trimble's workflow of re-adjusting until the test passes. ^[9] A failed test, by itself, doesn't automatically make an adjustment unusable, but it is never something to ignore. Before any adjustment is retained as final with a failed Chi Square test:

- **Investigate first.** Work the outliers and residuals as above.
- **Understand why it failed.** Is it a blunder, a systematic error, or standard errors that don't match the equipment and procedures?
- **Verify the residuals and independent checks.** No unexplained outliers, reasonable residuals, and agreement with check shots or known values.
- **Document a technical explanation** of why the result is still acceptable, including any weighting change, in the survey narrative.
- **Get review and approval** from the responsible PLS or survey lead.

■ 13 · NETWORK ADJUSTMENT

Network: fully constrained adjustment

Fully constrained = Does my survey agree with the project control?

Once the minimally constrained network is clean, hold the rest of the trusted project control. Because the observations have already been checked, a problem that appears only after the additional control is constrained strongly suggests a control, datum, transformation, or constraint issue. Still recheck the observations involved before concluding that the control is bad.

1 Add the remaining trusted control

On the constraints tab, check the fix boxes and, where appropriate, enter the standard errors for each additional control point. ^[9] Trimble's workflow fixes the horizontal (2D) control first to get an acceptable horizontal solution, then adds the elevation or height control. ^[9] On a difficult network, adding one control point at a time makes a bad one easy to spot.

2 Re-run the adjustment

Click **Adjust** and review the Results tab and the report again. ^[9]

3 Review the Control Coordinate Comparisons carefully

This report section shows the difference between each control point's known coordinates and the adjusted position the survey supports. ^[17] Compare these with the minimally constrained run.

Investigate a control point that stands out

If one control point disagrees significantly more than the rest, the problem is more likely the **control** than the survey observations: a bad published value, the wrong point identified in the field, or a datum or coordinate problem (for example a different datum realization or epoch, the wrong geoid, or grid vs. ground). Don't hold it until it's resolved. Leave it unfixed, report its comparison, and raise it with the PLS / survey lead.

Network: read the report

The report isn't just a pass/fail check. Trimble describes it as the tool for deciding which observations to disable, which control to fix, and which settings to change before you re-adjust. ^[17] Open it with the **Network Adjustment Report** button on the Adjust Network pane toolbar. It opens in your web browser. ^[9]

REPORT SECTION	WHAT IT TELLS YOU	WHAT TO LOOK FOR
Adjustment Statistics	Number of iterations, whether the adjustment passed or failed and why, the Reference Factor, the Chi Square test (95%), and degrees of freedom ^[17] ^[18]	A Reference Factor near 1.00. Degrees of freedom above zero; with zero there's no redundancy, so nothing is being checked. ^[18]
Adjusted Observations (GNSS, total station, leveling)	Each observation's standard error, residual (how much it had to be adjusted), and standardized residual, sorted worst first ^[17]	Observations in red failed the Tau test and are outliers. Trimble says to examine each one and justify keeping it. ^[17] ^[18]
Control Coordinate Comparisons	The difference between each control point's known coordinates and its adjusted coordinates ^[17]	One control point with a much larger difference than the rest points to bad control, the wrong point, or a datum or coordinate problem. Compare the minimally and fully constrained runs.
Adjusted Grid Coordinates	Adjusted northing, easting, and elevation, with their standard errors ^[17]	Coordinates and standard errors are reasonable for the survey; no point is much weaker than its neighbors.
Adjusted Geodetic Coordinates	Final adjusted positions. The Fixed column shows which coordinates were held. ^[17]	Only the control you intended to hold should be fixed.
Histogram of Standardized Residuals	The distribution of the normalized (standardized) residuals, with the critical Tau boundaries marked ^[17]	Most residuals should cluster near zero; anything beyond the Tau boundaries is a potential outlier to investigate.
Error ellipses [†]	The size and orientation of each point's positional uncertainty ^[17]	Large or very elongated ellipses mean weak geometry, such as a point tied in by only one observation.

For what each number means, see [What the numbers mean](#). For how to work through problems, see [Network: troubleshooting](#).

Network: save & report

1 Save the results by clicking OK

When you're satisfied with the adjustment, click **OK** to close Adjust Network. If you exit the command without clicking OK, the adjustment results are **not saved**. ^[9]

2 Know how TBC stores them

TBC saves the adjusted values as new coordinate records with quality *Adjusted* or *Fixed in network adjustment*, and promotes them to each point's final value. ^[9]

3 Re-adjust after later changes

If you change the project after adjusting, TBC asks whether to keep or clear the network adjustment. If you keep it, the results may no longer match the data, so re-run the adjustment once your changes are done. ^[9]

4 Save the Network Adjustment Report with the survey QA records

The report opens in your web browser. ^[9] Save it from the browser, for example as PDF, with the final run's date and job number, and file it with the survey QA records.

5 Review and record

Before sign-off, confirm the QA record covers:

- Reference Factor, Chi Square result, and degrees of freedom
- Residuals and standardized residuals, including any Tau outliers and how each was resolved
- Fixed coordinates (what was held, and how tightly)
- Adjusted coordinates
- Control Coordinate Comparisons
- Error ellipses and the confidence level they're shown at
- Disabled observations, with the reason for each
- Any weighting changes, with the reason

CLEAR ADJUSTMENT RESULTS CLEARS MORE THAN COORDINATES

Survey > Network > Clear Adjustment Results ^[9] doesn't just remove the adjusted coordinates. It also clears the adjustment transformation parameters, the coordinate fixes and adjustment flags, resets the weighting scalars to 1.00, removes the error ellipses, and recomputes the project. After clearing, you have to re-select the fixed control, re-enter any weighting, and re-run the adjustment. Save the report before you clear anything.

Reports & QA

Only the final adjustment should be represented as the final solution. A preliminary Traverse Adjustment Report may be retained as QA documentation of the traverse closure, if appropriate, but label it as preliminary. Remember that performing a network adjustment causes TBC to clear the traverse-adjusted coordinates and use the raw traverse observations, ^[21] so once the network adjustment is run, its report is the final one.

- **Traverse Adjustment Report** (traverse method): misclosures and precision before and after adjustment, end-point orientations, fixed points, and station adjustments. ^[3]
- **Network Adjustment Report** (network method): adjustment statistics, adjusted observations and residuals, control coordinate comparisons, adjusted coordinates, and error ellipses. ^[17] See [Network: save & report](#) for what the QA record must include.
- **Point Derivation Report**[†]: confirms which coordinate record each point is using.
- Export the final coordinates with **Home > Export**[†] using the office's standard PNEZD CSV format. In the survey narrative, record the adjustment method, the key results (precision ratio for a traverse; Reference Factor, Chi Square, and degrees of freedom for a network), and any disabled observations, weighting changes, or professional judgments.

Troubleshooting

SYMPTOM	LIKELY CAUSE	FIX
Angular misclosure is huge (degrees)	Wrong backsight point, or an orientation error at setup	Check the backsight ID and azimuth on each Station Setup
Misclosure is roughly one leg length	A station was skipped or selected out of order in Adjust Traverse	Edit the station list in Adjust Traverse so the stations are in occupation order ^[1]
Error grows with distance (scale-like)	Grid/ground mix-up, wrong combined scale factor, or ppm not applied	Recheck the Project Settings computation and scale settings, and the setup atmospherics
Elevation misclosure only	Wrong HI/HT, or a prism height changed without being recorded	Compare to field notes and edit the heights in properties
Every station gets adjusted, or closure looks meaningless	Neither end station is fixed. TBC treats a traverse endpoint as fixed only when its position and orientation are known independently of the traverse; Control quality alone doesn't make it fixed ^[2]	Verify each endpoint's station setup, control coordinate, backsight or foresight, and endpoint orientation in Adjust Traverse and in the Traverse Adjustment Report's End-Point Orientations and Fixed points sections ^[1] ^[3]
Traverse adjustment disappeared	A point in a manually adjusted traverse was merged ^[5]	Re-run Adjust Traverse and regenerate the report
Reference Factor well above 1 and Chi Square fails high	A blunder, or standard errors that are too optimistic (for example centering or height errors at 0.00)	Work the worst standardized residuals first; then check the default standard errors ^[16] ^[20]
Reference Factor below 1 and Chi Square fails low	Standard errors more pessimistic than the actual survey performance	Confirm the observations are sound, then consider revising the weighting with a documented reason ^[9]
Degrees of freedom = 0	No redundancy in the network	The statistics can't check the observations. Add redundancy, or use Adjust Traverse if it's a simple traverse.

SYMPTOM	LIKELY CAUSE	FIX
One control point disagrees much more than the rest	Bad control value, wrong point identified, or a datum or coordinate problem	Unfix it, report its comparison, and resolve it before holding it ^[17]
Network adjustment results disappeared	The command was closed without clicking OK, or results were cleared	Re-run Adjust Network and click OK to save ^[9]
Sideshots didn't move	The shots came from a different setup of the same point	Merge the setups, or include that setup in the traverse

■ 18 · SIGN-OFF

Checklist

Shared

- ☐ Coordinate system, geoid, units, and grid/ground confirmed
- ☐ Default standard errors set for total station and GNSS (none left at 0.00)
- ☐ Field data imported; plan view looks right
- ☐ Control points set to Control quality, with IDs matching
- ☐ HI/HT, prism constants, and atmospheric checked
- ☐ Final coordinate solution confirmed (traverse or network adjustment), and the final report retained is identified; any preliminary report is labeled as QA only

Traverse adjustment

- ☐ Traverse created; angular and linear closure within spec
- ☐ Traverse adjusted; Traverse Adjustment Report reviewed and saved

Network adjustment

- ☐ Network Adjustment settings and default standard errors reviewed

- ☐ Minimally constrained adjustment run; outliers investigated and resolved

- ☐ Degrees of freedom above 0; Reference Factor reviewed; Chi Square passes, or a failure is documented and approved by the PLS / survey lead

- ☐ Fully constrained adjustment run; Control Coordinate Comparisons reviewed

- ☐ Results saved with OK; Network Adjustment Report saved with QA records

- ☐ Disabled observations and any weighting changes documented with reasons

Sign-off

- ☐ Final coordinates exported and adjustment method and results noted in the narrative

- ☐ Reviewed by PLS / survey lead

■ 19 • SOURCES

References

Trimble help pages were checked in October 2026. Trimble updates its help with each TBC release, so confirm them against the version you have installed.

TRIMBLE BUSINESS CENTER HELP

- [1] Trimble. "Create, Edit, and Adjust a Traverse." *Trimble Business Center Help*.
<https://help.fieldsystems.trimble.com/tbc/1927.htm>

- [2] Trimble. "Understanding Traverses." *Trimble Business Center Help*.
<https://help.fieldsystems.trimble.com/tbc/16947.htm>

- [3] Trimble. "Run a Traverse Adjustment Report." *Trimble Business Center Help*.
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- [4] Trimble. "traverse adjustment" (glossary). *Trimble Business Center Help*.
<https://help.fieldsystems.trimble.com/tbc/3013.htm>

- [5] Trimble. "Computation Settings." *Trimble Business Center Help*.
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<https://help.fieldsystems.trimble.com/tbc/10262.htm>
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- [7] Trimble. "Understanding Point Types." *Trimble Business Center Help*.
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- [8] Johnson, E. "TIP #148: How to Change the Quality of Multiple Points at Once." *Trimble Community*, 2021.
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TRIMBLE FIELD SOFTWARE (FOR COMPARISON ONLY; NOT CITED FOR TBC OPTION NAMES)

- [10] Trimble. "Traverse calculations." *Trimble Access Help*. help.fieldsystems.trimble.com/trimble-access/latest/en/cogo-traverse.htm. See also "Traverse (Optional Module)," *Trimble FieldLink Help*:
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TEXTBOOKS & STANDARDS

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- [12] Ghilani, C. D. *Adjustment Computations: Spatial Data Analysis*, 6th ed. Wiley, 2017. Least-squares traverse and network adjustment, reference variance, χ^2 test, and blunder detection.
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- [13] Washington State Department of Transportation. *Highway Surveying Manual*, M 22-97 (current revision). Survey accuracy standards and closure requirements for WSDOT work.
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- [14] NSPS / ALTA. *Minimum Standard Detail Requirements for ALTA/NSPS Land Title Surveys* (current edition). Relative positional precision requirements.
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- [15] Mahun, J. "Adjustment Methods Comparison" (Compass, Transit, Crandall). *Open Access Surveying Library*.
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- [16] Blecha, J. "Tip of the Week #29: Sources of Default Standard Errors." *Trimble Community*, 2020.
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TRIMBLE BUSINESS CENTER HELP (NETWORK ADJUSTMENT)

- [17] Trimble. "Run a Network Adjustment Report." *Trimble Business Center Help*.
https://help.fieldsystems.trimble.com/tbc/398_1.htm
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- [19] Trimble. "Network Adjustment Settings." *Trimble Business Center Help*.
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- [20] Trimble. "Workflow: Import Total Station Data." *Trimble Business Center Help*.
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Parametrix

Internal training material · Traverse & Network Adjustment in TBC · Verify menu paths against your installed TBC version